MILITARY SPECIFICATION

TRANSPORTER: M945 CHASSIS, FOR RIBBON FLOATING BRIDGE

This specification is approved for use by the USA Belvoir Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers modifications to the M945 5-ton truck chassis and equipment to be installed on the chassis to convert the chassis to a ribbon bridge transporter that is capable of launching, retreiving and transporting all components of the ribbon bridge system.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

UU-T-81	- Tags, Shipping and Stock.
PPP-P-40	- Packaging and Packing of Hand Tools.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-R-196	- Repair Parts, Accessories, and Kits, Mechanical; Packaging of.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-E-10062	- Engines: Preparation for Shipment and Storage of.
MIL-G-20241	- Gasket Material, Wood Felt, Impregnated, Adhesive, Pressure Sensitive.
MIL-L-21260	- Lubricating Oil, Internal Combustion Engine, Preservative and Break-In.
MIL-T-22085	- Tape, Presure Sensitive, Adhesive, Preservation and Sealing.
MIL-P-23377	- Primer Coating, Epoxy Polyamid, Chemical and Solvent Resistant.
MIL-A-46153	- Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty, Single Package.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-C-81346	- Compound, Deck Covering, Nonslip, Lightweight, for Aluminum Alloy Landing Mats.
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-281	- Automobiles, Trucks, Truck-Tractors, Trailers and Trailer Dollies; Preservation and Packaging of.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.
MIL-STD-1261	- Welding Procedures for Constructional Steels.

DRAWINGS

ME

TA 13226E6250	- Transporter, Ribbon Bridge, M945 Chassis.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN WELDING SOCIETY, INC. (AWS)

D1.1 Structural Welding Code, Section 5, Qualification.

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(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The ribbon bridge transporter or separately procured component(s) thereof shall be in accordance with TA13226E6250 or the applicable drawing(s) and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the requirements shown on the drawings is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly and operation of the transporter. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article. When specified (see 6.2) the ribbon bridge transporter(s) or separately procured component(s) shall be subjected to first article inspection and test (see section 4 and section 6).

3.2.1 Hydraulic system components. In addition to the preproduction model, the following hydraulic components shall be furnished and tested by the contractor prior to the time the preproduction model is furnished: One each of all pressure relief and check valves, flow regulators, quick disconnect couplings, and one valve control bank. The hydraulic components shall be identical to those furnished on the preproduction model. Tests shall be those specified in section 4 and shall be subject to surveillance and approval by the Government.

3.2.2 Transportation data. When specified (see 6.2), a preproduction model transporter shall be weighed, measured, and the center of gravity located to provide the data required for the transportation data plate.

3.3 Material. Material shall be as specified herein and on the applicable drawings. Material not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Copies of analyses and test reports for all materials shall be available for inspection. Chemical composition and mechanical properties of heat-treated materials shall be verified by chemical analysis and mechanical tests. Unless otherwise specified (see 6.2), the age of rubber products furnished shall not exceed 24 months from the date of issue to date of acceptance of the transporter or separately procured component(s) by the Government.

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3.3.1 Material deterioration and control. The transporter or separately procured components shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environments to which the item may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by an insulating barrier such as a coat of epoxy primer conforming to MIL-P-23377, or a corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241. Protection against any galvanic corrosion could also be obtained by exclusion of the electrolyte if feasible.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and subcomponents, and shall make information available, upon request, to the contracting officer or designated representative.

3.4 Structural performance. The transporter shall perform all tests specified herein without structural failure or permanent deformation attributable to nonconforming materials, dimensions, incorrect assembly or adjustment, tolerances, or incorrect workmanship.

3.5 Fuel, coolant, lubricants, and hydraulic fluid. The transporter shall be serviced, tested and furnished with standard military service products of the types and grades as specified on the lubrication order (LO). The hydraulic system shall be filled and tested with oil conforming to MIL-L-2104, grade 10, and shall not show visual evidence of contamination when examined as specified herein.

3.6 Transporter assembly. The transporter shall be assembled utilizing the Government-furnished 5-ton, 6 x 6, M945 truck chassis conforming to top assembly TA13226E6250 (see 6.6). All other items shall be fabricated and installed or procured and installed by the contracor.

3.6.1 Transporter and transporter systems assembly, installation and adjustments. The components of the various systems of the transporter shall be assembled, installed and adjusted to assure acceptable performance of the completed transporter as specified herein. All like components shall be interchangeable with respect to installation, performance and quality. All adjustments not requiring an operational transporter shall be completed prior to initial start-up. Care shall be exercised to assure that final adjustments which must be completed after the transporter is operational will not result in a safety hazard when the engine is started.

3.6.2 Initial start-up. Prior to starting the engine with power take-off engaged, the hydraulic system shall conform to the cleanliness requirements specified in 3.8.2, and the entire transporter shall be serviced in accordance with 3.5. The engine shall be prepared, started and operated in accordance with the engine manufacturer's run-in and operating procedures. Upon starting the engine, all final adjustments shall be performed and examined as specified in 4.5 to assure correct assembly and operation.

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3.7 Electrical system. The electrical circuitry and connection shall be assembled and installed to assure conductance of the full electrical capacity of the components connected. The full electrical insulation qualities of all dielectric components shall not be reduced by assembly or installation. Electrical connection points shall be bare and clear of paint and other dielectric materials. Electrical cables shall be installed without excess slack and secured to prevent slippage through the cable clamps. The electrical circuits shall operate their respective electrical functions when tested as specified in 4.5.2.2.

3.8 Hydraulic system.

3.8.1 Hydraulic system performance. The hydraulic system shall be assembled, installed, cleaned, serviced with the specified fluid, bled of air, and adjusted to assure acceptable performance of each hydraulic function. The hydraulic pump shall provide the fluid power, in accordance with the applicable drawings, to all hydraulic functions and withstand the relief valve pressure setting without evidence of permanent deformation or external leakage. The controls and components of the hydraulic circuitry shall start, correctly operate, and stop all hydraulic functions in a smooth controlled fashion without binding, chatter, erratic response, vibration, or abnormal noise. The hydraulic pump shall develop a minimum of 2400 psi at 1000 engine revolutions per minute (rpm) when the hydraulic oil temperature is 180° F, ±5° F.

3.8.2 Hydraulic system cleanliness. Prior to their installation into the transporter, the interior and exterior surfaces of all hydraulic components shall be washed and flushed throughout to remove all visible contaminants, particles and foreign elements. Openings of the components shall be immediately covered to prevent the entry of contaminants and shall remain covered until installation. Prior to initial start-up, the assembled hydraulic system, without the pump, reservoir, winch motor, boom cylinder, and filter elements, shall be flushed throughout utilizing an auxiliary pump, reservoir, and filtration arrangement. After flushing, the hydraulic fluid shall meet the contamination limits specified in table I. The remaining components shall then be installed and the system, after being filled with the specified hydraulic fluid, shall be operated and bled of air prior to being subjected to the 10-minute operational test described in 4.5.2.4. Upon completion of all tests and prior to final acceptance, the hydraulic filter element shall be replaced and the hydraulic fluid shall meet the cleanliness requirements specified in table I.

TABLE I. Contamination limits (particles per milliliter).

Quantity (max)	Particle size
1000	Greater than 10 microns
100	" " 20 "
40	" " 30 "
10	" " 40 "

3.8.3 Hydraulic system installation. Tubing and components shall be as specified herein and on the applicable drawings. The components and tubing

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shall be painted as specified in 3.9 prior to installation and shall be secured by brackets or clamps to prevent movement or chafing of the tubes. Connection points shall be bare and clear of paint, and special care shall be taken to prohibit the entrance of the sealant into the system.

3.9 Treatment and painting. Cleaning, treatment and painting of components shall be in accordance with the applicable drawings, and as specified herein. Where MIL-T-704, type B, paint is required, the finish coat, shall conform to MIL-C-46168, camouflage green 383. Surfaces inaccessible for painting after installation shall be painted before installation. All surfaces, with the exception of the winch cable, glass, rubber, bearing surfaces, and the piston rod, shall be painted. Nonslip coating conforming to MIL-C-81346 shall be applied to all walkway surfaces, the top surface of the front support, and the top surface of the boom assembly, as observed in the folded or down positon.

3.10 Marking. Identification markings, level indications, jacking and lifting points, warnings, and all other marking shall be as specified on the applicable drawings.

3.11 Government-loaned property. When specified (see 6.2), any or all of the following property in the quantities indicated will be loaned by the Government (see 6.5):

Item	Quantity	Description	Identification
1	1	Bay Assembly, Interior	5420-00-071-5322, type I
2	1	Bay Assembly, Ramp	5420-00-497-5276, type II
3	1	Sling, Bay Lifting	3940-00-214-7493
4	1	Snatch Block	3940-00-338-5653
5	1	Operator's Manual	TM5-5420-209-12

3.12 Performance. The transporter shall perform as specified herein without evidence of any of the following defects:

a. Malfunctioning of components.
b. Binding of parts or improper fit during launching and retrieving of bridge bays.
c. Permanent distortions (other than normal wear).
d. Material defects such as cracks and open joints.
e. Failure of mechanical mechanisms to operate as required.
f. External hydraulic leakage.
g. Failure of hydraulic systems to operate as required.
h. Failure of winch to operate as required.

3.12.1 Operational characterisitcs. The transporter shall:

a. Be capable of launching a bridge bay within 5 minutes and retrieving it within 10 minutes.
b. Perform "high-bank" launches and retrievals as described in the approved technical manual (TM).

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3.12.2 Equipment publications. Operating and maintenance functions of the transporter shall conform to applicable sections of the approved technical manuals (TM) (see 3.12, 6.7, and 6.8).

3.13 Reliability. Each trasnporter shall have a mean missions between failure of 50 missions. A mission consists of the required transport distance and one launch/retrieve cycle (see 4.5.2.7.1).

3.14 Maintainability. The transporters shall have a maintenance ratio of no more than 0.03 man-hours of maintenance per hour of operation when operated and maintained in accordance with the approved technical manuals and when tested as specified in 4.5.2.7.

3.15 Interchangeability. All parts, components and assemblies having the same identification number shall be functionally and dimensionally interchangeable. Interchangeability is defined as two or more parts (components and assemblies) possessing such functional and physical characteristics as to be equivalent in performance and durability and capable of being exchanged one for the other without alteration of themselves or of adjoining parts (components and assemblies), except for built-in adjustment, and without selection for fit and performance.

3.16 Specification-control items and source-control items. Prior to using any specification-control item other than a suggested source item, the contractor shall furnish to the contracting officer evidence that the item conforms to all requirements on the applicable drawing. The contractor is responsible for performing all inspections and tests on specification or source control items if not performed by the vendor. No change shall be made to any source or specification control item after approval of the preproduction model without prior approval by the contracting officer.

3.17 Workmanship.

3.17.1 Transporter assembly. The transporter shall be assembled in accordance with the requirements specified on the applicable drawings and specifications. All parts, components and assemblies containing moving parts shall be assembled free of harmful extraneous materials such as metallic particles, dirt, grit, or corrosion.

3.17.2 Castings and forgings. Castings and forgings which are components of the transporter shall be supplied free of harmful extraneous material such as sand, dirt, pits, sprues, scale, and flux. Rework shall be limited to procedures which do not reduce strength or affect function.

3.17.3 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing for steel only. All bends shall be made with

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controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. All parts shall be completely and accurately shaped to the drawing requirements prior to joining (welding). Joints, butts and splices shall be limited to those shown on the end product drawings. External surfaces shall be free of burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.17.4 Steel welds. The weld types, materials, sizes, and quality of soundness shall conform to the weld requirements specified on the applicable drawings. Prior to the production fabrication of any steel weldment, the contractor shall prepare welding procedures and workmanship specimens in accordance with the requirements of MIL-STD-1261 and shall submit such procedures and specimens to the Government. Production welding shall be permitted only after submitted procedures and specimens have been approved by the Government. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall transmit stress without permanent deformation of failure when the parts connected by the welds are subjected to proof and service loading. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 3/16 inch unless appropriate bridging techniques are used.

3.17.5 Care of welding electrodes. Care of welding electrodes specified for welding high-strength steel shall be as follows:

a. When a container of welding electrodes is opened, the entire contents shall be placed immediately in a baking oven, with electrodes placed not more than three layers deep on a tray, and baked either (1) for a minimum of 1 hour, but not more than 1-1/4 hours, at 800° F; or (2) for a minimum of 4 hours at 400° F.
b. The baked electrodes, while still warm, shall then be placed in a ventilated holding oven set at 250° F to 300° F.
c. Not more than 1/2-hour supply of welding electrodes shall be taken from the holding oven at one time.
d. If electrodes are out of the holding oven in excess of 1/2 hour but less than 1 hour, they shall be returned to the holding oven for a period of not less than 1 hour.
e. If electrodes are out of the holding oven in excess of 1 hour, they shall be returned to the baking oven and rebaked as described under "a".
f. Electrodes removed for use shall not be allowed to come in contact with any moist or wet surfaces. Electrodes that have been in contact with water, grease or oil shall not be used.

3.17.6 Welders. Before assigning any welder to fabricate transporters covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by AWS D1.1, Structural Welding Code, Section 5, Qualification, for the type of welding operations to be performed and that such qualification is effective as defined by the code. Test specimens shall be made on the same alloy, gage and temper of material with the same filter materials and in the

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same position as will be used in production. Welders who have been similarly qualified under other recognized codes may be considered qualified for the purposes of this specification, provided they are approved by the contracting officer.

3.17.7 Weld repairs. All areas of defective or unacceptable welds shall be repaired in accordance with the weld repair procedures specified in MIL-STD-1261 and shall be subject to prior approval of the contracting officer. Repair of any defective weld shall be limited to two rewelds.

3.17.8 Hoses and tubing. All hoses and tubing shall be installed without twists, kinks or collapsed sections. Tubing bends shall be in accordance with the drawing requirements. Tubing shall not be reformed. Tubing assemblies shall be correctly formed to prevent prestress at installation due to misalinement. Tubes, hoses or their connections shall not leak when tested as specified in 4.5.2.

3.17.9 Machine work. Tolerances and gages for metal fits shall conform to the limits specified on the applicable end product drawings.

3.17.10 Jigs and fixtures. Fabrication, assembly and installation of all transporter components (except minor parts) shall be accomplished with production tooling, jigs or fixtures, and parts shall be joined while held in position. Jigs or fixtures shall be designed to minimize distortion of the parts being welded and to assure that the configuration, alinement and location of parts meet the specified requirements. Subsequent straightening and thermal stress relieving is permitted to relieve distortion provided the straightening process does not damage the component.

3.17.11 Bolt torque. Unless otherwise recommended by the component manufacturer, bolts and screws shall be installed in accordance with the torque values specified on the drawings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined and tested in accordance with referenced specifications, standards and drawings, as applicable.

4.1.2 Disassembly inspection. Failure of any examination or test by the first article transporter(s) or separately procured component(s) shall be cause

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for disassembly, in the presence of a Government representative, of the preproduction model to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the preproduction model. Reassembly with replacement parts and retesting shall be the responsibility of the contractor.

4.1.3 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the quality assurance provisions (QAP) shown on the drawings. The drawings specify the characteristics requiring QAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.8 and 6.9).

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection comparison (see 4.6).
d. Inspection of packaging (see 4.7).

4.3 First article inspection.

4.3.1 Examination. The preproduction transporter or separately procured component(s) shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article transporter(s) shall be subjected to the tests specified in 4.5.2.1 through 4.5.2.7. Failure of any test shall be cause for performing the disassembly inspection specified in 4.1.2.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests for separately procured transporter components shall be in accordance with MIL-STD-105, inspection level III.

4.4.2 Examination.

4.4.2.1 Individual. Each transporter or separately procured component(s) shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2.2 Samples. Samples of separately procured components selected in accordance with 4.4.1 shall be examined for the applicable defects specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 6.5 percent defective for minor defects.

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4.4.3 Tests.

4.4.3.1 Individual. Each transporter shall be tested as specified in 4.5.2.1 through 4.5.2.6. Failure of any test shall be cause for rejection.

4.4.3.2 Samples. Samples of separately procured components selected in accordance with 4.4.1 shall be subjected to the applicable test(s) specified in 4.5.2.1 through 4.5.2.6. AQL shall be 2.5 percent defective.

4.5 Inspection schedule.

4.5.1 Examination. The transporter(s) or separately procured component(s) shall be examined as specified herein for the following defects, as applicable:

Major

101. Any part or component not in accordance with the QAP requirements as shown on the end product drawings.
102. Any dimension other than those identified on the QAP requirements shown on the drawings not as specified.
103. Transportation data not as specified, when required.
104. Materials not as specified.
105. Materials not corrosion resistant or treated to be made corrosion resistant for applicable storage and operating environment.
106. Dissimilar metals not effectively insulated from each other.
107. Identification of material finishes not documented as specified.
108. Structural performance not as specified.
109. Fuel, coolant, lubricants, and hydraulic fluid not as specified.
110. Transporter and transporter systems assembly, installation and adjustments not as specified.
111. Hydraulic system performance and installation not as specified.
112. Cleaning, treatment and painting not as specified.
113. Identification marking missing or not as specified.
114. Transporter performance not as specified.
115. Operational characteristics not as specified.
116. Components not interchangeable as specified.
117. Specification and source control items not as specified.
118. Workmanship not as specified.
119. Welder qualification not as specified.
120. Age of rubber products not within specified limits.
121. Torque of bolts not as specified.
122. Components missing or not as specified.
123. Nonskid compound missing or not as specified.
124. Lubrication fittings not as specified.

4.5.1.1 Welding examination. All areas of welded assemblies affected by the welding processes shall be examined prior to plating, painting, or installation for evidence of defects, including base metal cracks, weld cracks, under cutting, lack of fusion, porosity, lack of penetration, incorrect fit, incorrect preparation for welding, inclusions, incorrect weld size, and incorrect welding

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procedures. Weld quality shall be examined and compared against the approved welding specimens (see 3.17.4). All welds shall be visually examined with the aid of a 5X magnifying glass. Evidence of weld defects in excess of the soundness requirements shown on the applicable drawings shall constitute failure of this examination.

4.5.1.2 Weld repair examination. All weld repair areas shall be visually examined with the aid of a 5X magnifying glass. Weld repairs which evidence defects in excess of that exhibited by the weld specimens or soundness requirements specified in 3.17.7 shall constitute failure of this examination.

4.5.2 Tests.

4.5.2.1 Test conditions.

a. All test apparatus shall be accurate to within 3 percent of full-scale reading. The transporter instruments may be used to the extent possible for test data readings. All pertinent test data shall be recorded and validated by the contractor and by the Government representative.
b. The transporter shall be serviced in accordance with the lubrication order (LO) as contained in the TM (see 3.12.2). Fuel and lubricants shall be those designated for use in the ambient temperature at the place of test.
c. Unless otherwise specified (see 6.2), the transporter shall be equipped with an interior bay (see 3.11), and all tests shall be performed at the climatic conditions existing at the place of test.
d. The transporter shall operate as specified in 4.5.2.2 through 4.5.2.6 without maintenance other than that established by the lubrication order contained in the approved technical manual (TM).
e. During the test, all equipment shall be operated in accordance with the approved TM.
f. The sequence of testing shall be as follows:

(1) Electrical system circuitry. (Sequence optional)
(2) Hydraulic system pretest cleanliness.
(3) Hydraulic system performance.
(4) Lift and loading test.
(5) Hydraulic system posttest cleanliness.
(6) Reliability and maintainability.

4.5.2.2 Electrical circuitry. Actuate each electrical circuit of the transporter and observe the response of each respective electrical function. Nonconformance to any drawing requirement or to any specification referenced herein shall constitute failure of this test.

4.5.2.3 Pretest flushing and cleanliness. Verify that the hydraulic system has been flushed in accordance with 3.8.2 and take a particle count. If the particle count exceeds the values in table I, continue flushing and sampling until contamination is reduced to the acceptable level. After the specified cleanliness is reached, replace the filter element, connect the pump, reservoir,

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winch, and boom cylinder into the system and fill with oil conforming to MIL-L-2104, grade 10. Nonconformance to the requirements of table I shall constitute failure of this test.

4.5.2.4 Hydraulic system performance. Operate the system for 10 minutes, actuating the boom and winch in both directions. At the end of this 10-minute period set the engine at 1000 rpm and actuate the boom control valve by shifting the lever to the down position. Record the relief valve setting for a sustained 3-minute period. Continue to operate against the relief valve until the oil temperature reaches 180° F, ±5° F. Verify that the pump will produce 2400 psi at 1000 engine rpm when the oil temperature is 180° F, ±5° F. Record maximum pressure developed when working against the relief valve at an engine speed of 1900 rpm. Operate the boom and the winch sufficiently to detect any chatter, vibration or erratic response. Failure to function smoothly or to develop the required pressure shall constitute failure of this test.

4.5.2.5 Lifting and loading performance. Place the bogie brackets in position and attach the boom cable to the lifting sling for a controlled high-bank launch (see approved technical manual (TM)). Lift an interior bay assembly and hold approximately 1 foot high for 5 minutes. Lower the load approximately 6 inches and hold for 3 minutes. After successful completion of this lift, lower the load, re-rig, remove bogie brackets, and load the bay on the transporter. Verify that the travel restraints and loading brackets fit and function correctly. Failure to hold the load, to function, or to fit as prescribed on the drawings shall constitute failure of this test. Remove the transporter pintle hook during this test.

4.5.2.6 Posttest hydraulic cleanliness. After completion of the above tests, the hydraulic system cleanliness shall be checked by taking a particle count. If the particle count exceeds the values in table I, the system shall be flushed until contamination is reduced to the acceptable level. Inability to conform to the cleanliness levels specified in table I shall constitute failure of this test. After achieving the required cleanliness, and prior to delivery to the Government, a new filter element shall be installed.

4.5.2.7 Reliability and maintainability. This test will be conducted by the Government at a site selected by the Government. The test will be conducted using a minimum of 14 transporters. Maintenance shall be performed in accordance with the technical manual, and at the end of each day's mission, any damage to the transporters will be repaired prior to the next day's mission. Failure to meet all performance requirements of this specification, when such failure is due to faulty material or workmanship, shall be cause for refusal by the Government to continue acceptance of production units until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on all units previously produced. Any such material or workmanship deficiencies will be considered prima facie evidence that all units accepted prior to the completion of this test are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

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4.5.2.7.1 Reliability-maintainability test cycle. Reliability-maintainability testing shall be conducted for 50 bridge missions for each transporter. Each bridge mission for a transporter shall consist of the following:

a. 1 launch/retrieve cycle.
b. 20 miles of transporting.

4.5.2.7.2 Reliability. Test each transporter for 30 bridge missions. If zero or one failures occur for each individual transporter, reliability is acceptable. If two failures occur for any of the transporters, continue testing those transporters with two failures until 100 missions are complete. If no more than two failures occur for each transporter, reliability is acceptable. If at any time during the test three failures occur for any transporter, reliability is not acceptable and the test is terminated. For the purpose of reliability computations a mission failure is defined as follows:

a. Any malfunction which would cause inability to complete a bridge mission specified in 4.5.2.7.1 and cannot be repaired by the crew or organizational maintenance within one hour. Any malfunction requiring repair at a higher maintenance level is considered a mission failure.
b. Any impending malfunction which would require maintenance that cannot be deferred until the end of the bridge mission which would create a safety hazard if the mission continued and cannot be repaired by the crew or organization maintenance within one hour.

Simultaneous related malfunctions are considered one failure.

4.5.2.7.3 Maintainability. The maintenance manhours required to perform all corrective and preventive maintenance occuring during the reliability test (see 4.5.2.7.1) shall be recorded and classified as to whether it was scheduled or unscheduled. The MR requirement specified in 3.14 shall be assessed using the following formula:

$$MR = \frac{\text{total scheduled maint manhrs + total unscheduled maint manhrs}}{\text{total operating time}}$$

Operator/crew checks and services are excluded from this ratio.

4.6 Inspection comparison. The Government may select transporters at any time during the contract production period and subject these transporters to the examination specified in 4.5.1 and the tests specified in 4.5.2 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on transporters selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production transporters. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

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4.6.1 Inspection failure. Failure of an inspection comparison transporter to meet any requirement specified herein during and as a result of the examination and tests specified in 4.6 shall be cause for rejection of the inspection comparison transporter and shall be cause for refusal by the Government to continue acceptance of production transporters until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies.

4.7 Inspection of packaging. The preservation and marking of each transporter as prepared for shipment shall be examined to assure conformance to the requirements of section 5 of this specification and the quality assurance provisions of MIL-STD-281 and MIL-E-10062.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or B or commercial as specified (see 6.2).

5.1.1 Level A. Each complete transporter shall be preserved in accordance with the level A (mobile) preservation and packaging requirements of MIL-STD-281, with the following exceptions and additional requirements. Materials and methods required for preservation and packaging specified herein shall conform to the applicable specifications listed in MIL-STD-281.

5.1.1.1 Preservation. Preservative oil specified in MIL-STD-281 as P-10, grade 1 or 2, shall conform to MIL-L-21260, grade 10 or 30, as applicable.

5.1.1.2 Engine. The transporter engine shall be preserved in accordance with the level A requirements of MIL-E-10062, type I.

5.1.1.3 Cooling system. The cooling system shall be preserved by the water and antifreeze procedures as specified in MIL-STD-281, except the antifreeze shall conform to MIL-A-46153.

5.1.1.4 Repair parts. Repair parts shall be preserved in accordance with the level A requirements of MIL-R-196. Parts not specifically covered in MIL-R-196 shall be preserved in accordance with the applicability, selection and application provisions of MIL-P-116.

5.1.1.5 Maintenance tools. Maintenance tools shall be preserved and packaged in accordance with the level A preservation and packaging requirements of PPP-P-40.

5.1.1.6 Batteries. Batteries shall be secured in place in the battery carriers as specified in MIL-STD-281.

5.1.1.7 Windshield cover. The windshield shall be protected from breakage by covering each side with a fitted piece of 1/4 inch thick plywood or hardboard allowing a minimum of 1 inch clearance between the glass and the cover. The cover shall be secured in place as specified for OVE equipment in MIL-STD-281.

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5.1.1.8 Hydraulic system (except hydraulic brake or steering systems). The hydraulic fluid supply tank shall be filled with the hydraulic fluid recommended for operation. The pistons shall be retracted as far as practicable into the cylinder and secured. When the pistons cannot be fully retracted, exposed portions of the piston rods (ramshafts) shall be coated with type P-6 or P-11 preservative and wrapped or covered with barrier material conforming to MIL-B-121, type I or II, grade A, class 2, extending the wrap approximately 2 inches on the ram cylinder, if possible. The wrap shall be secured in place with tape conforming to MIL-T-22085, type II. When the pistons can be fully retracted, any remaining uncoated surfaces of the piston rods shall be coated with type P-1 preservative, with no wrapping required. The hydraulic control valves shall be secured in the neutral position. A tag conforming to UU-T-81, type A, shall be attached in a conspicuous location indicating: "The hydraulic system is filled with the fluid recommended for operation." The information shall be applied on the tag with waterproof ink.

5.1.1.9 Other components. Other components, such as the boathook, tethering line, spanner wrench, wrecking bar, bay connecting wrench, and lanyard, shall be placed and secured as provided for on the transporter.

5.1.2 Level B. Each complete transporter shall be preserved in accordance with the level B (mobile) preservation and packaging requirements of MIL-STD-281, with the exceptions and additional requirements specified in 5.1.1 for level A.

5.1.3 Commercial. Each complete transporter, except for the engine and cooling system (see 5.1.3.1), shall be preserved in accordance with MIL-STD-1188.

5.1.3.1 Engine. The cooling system, fuel system, and combustion chambers and valves shall be preserved in accordance with MIL-E-10062, as specified for type I classification. The cooling system shall be preserved by the water and anti-freeze procedure using antifreeze conforming to MIL-A-46153.

5.2 Packing. Each complete transporter, preserved as specified in 5.1, shall not require additional packing (see 6.10).

5.3 Marking (see 6.10).

5.3.1 Military. Marking for military levels of protection (level A or B) shall be in accordance with MIL-STD-281.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The transporter is intended for use as an on-road or off-road overland carrier of ribbon bridge interior bays and ramp bays, the standard 27-foot bridge erection boat or ribbon bridge erection boat mounted on

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its boat cradle or cargo pallets. It is intended to operate with rear axles partially submerged in water while launching or retrieving the bridge bays or the boat.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Date of issue of DoDISS applicable and exceptions thereto (see 2.1.1).
c. When first article transporters or separately procured component(s) are required and the number of transporters or components required (see 3.2).
d. When data for the transportation data plate is required (see 3.2.2.).
e. Age of rubber products if other than as specified (see 3.3).
f. When Government-loaned property is required (see 3.11 and 6.5).
g. When an interior bay is not required for testing (see 4.5.2.1).
h. Degree of preservation required (see 5.1).

6.3 First article. When a first article inspection is required, the contractor shall furnish one or more transporters or separately procured components for evaluation and test within a specific time frame. Transporters or separately procured components to be tested may be either preproduction models or first-produced items. Examinations and tests shall be as specified in section 4. The contracting officer shall include specific instructions in the acquisition documents regarding arrangements for examinations specifying which tests will be conducted by the contractor, which tests if any will be conducted by the Government, the time frame for testing and submittal of the first article documentation, how the first will be monitored by the Government, and how the first article will be submitted and approved. Any changes or deviations of the transporter(s) will be subject to approval of the contracting officer. Approval of the first article does not relieve the contractor of his obligation to furnish transporters or separately procured components conforming to this specification.

6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operational manual to be furnished with each transporter.

6.5 Government-loaned property. The contracting officer shall arrange to loan the property specified in 3.11 in the quantities required.

6.6 Government-furnished property. The contracting officer should arrange to furnish the property specified in 3.6.

6.7 Provisioning. The contracting officer should include provisioning requirements, if required, for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of transporters or separately procured component(s).

6.8 Quality assurance provisions (QAP). The contracting officer should require that the contractor maintain records of all QAP inspections. A suggested paragraph is as follows:

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"The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include, as a minimum, lot size, sample size, drawing requirement, actual measurement, number and type of deficiencies found, quantity approved, quantity rejected and corrective action taken when applicable."

6.9 Definitions.

6.9.1 Quality assurance provisions (QAP). A QAP is a contractural requirement that supplements section 4 of the specification. QAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.10 Packaging. Degrees of packing have not been included as only mobile shipment is appropriate. For purposes of preservation/packing level marking, the level of packing shall be considered level A.

6.11 Storage prior to delivery. The contracting officer should arrange for the necessary storage provisions for transporters awaiting delivery to prevent damage or deterioration.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-53031(ME)	Transporter: M945 Chassis, for Ribbon Floating Bridge

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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